

Yuxin Wang

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EDUCATION

Johns Hopkins University 01/2026 - Present
Department of Applied Mathematics and Statistics
Major: M.S.E. in Data Science GPA: 3.58/4

Tongji University, Shanghai, CN 09/2019 - 06/2023
College of Electronics and Information Engineering
Major: Data Science and Big Data Technology GPA: 3.55/4

RESEARCH EXPERIENCE

Improving AutoResearch for SciML via Loss Landscapes and Spectral Diagnostics 07/2026 - Present
Advised by Yaoqing Yang and Pu Ren, Dartmouth College

- Partitioned FNO training states into regimes, identified optimization, approximation, and generalization bottlenecks, and classified them using loss landscapes, Hessian spectra, CKA, mode connectivity, and layer-wise weight and Jacobian Gram spectra.
- Studied transfer across model capacities by building trajectory zoos from reduced-capacity FNOs and retrieving relevant optimization histories for larger models.
- Designed diagnostic-informed skills and integrated them into the LLM AutoResearch pipeline to guide model-improvement proposals.
- Reduced relative L2 error from 0.016 to 0.003 after 10 research rounds, compared with 0.005 for AutoResearch without diagnostics.

Self-Evolving Medical Agent via Counterfactual Reasoning 03/2026 - 09/2026
Advised by Xing Han, Johns Hopkins University

- Implemented CF-Evolve, a Proposer-Verifier-Solver agent in which accepted counterfactuals accumulate across rounds as reusable context for a frozen Solver.
- Curated 3k counterfactual instructions; trained a Qwen2.5-7B Proposer with LoRA SFT and reinforcement learning (GRPO) using Solver/Verifier rewards.
- Evaluated self-evolution on MIMIC-IV clinical prediction and HoVer fact verification, including transfer to harder held-out cases.
- Improved MIMIC-IV IHM AUROC from 0.700 to 0.752 and reduced LOS MAE from 59.4 to 55.9 h while keeping the Solver frozen.

Unveiling Multi-Regime Patterns in SciML: Distinct Failure Modes and Regime-Specific Optimization 03/2025 - 05/2026
Advised by Yaoqing Yang and Pu Ren, Dartmouth College

- Developed mode-connectivity measurements for PINNs and characterized models into three regimes using Hessian, CKA, and mode-connectivity diagnostics.
- Implemented hard constraints and curriculum learning for PINNs, PINOs, and Neural ODEs to expose regime-specific optimization behavior.
- Identified SciML pathologies that differ from those observed in computer vision through loss-landscape analysis.

PUBLICATIONS

- Counterfactual Self-Evolving Agents for Evidence-Grounded Reasoning. Submitted to ICLR 2027 (under review).
- Unveiling Multi-Regime Patterns in SciML: Distinct Failure Modes and Regime-Specific Optimization. ICML 2026 (accepted).

PROFESSIONAL EXPERIENCE

Johns Hopkins University, Baltimore, USA 06/2026 - Present
Subject Matter Expert - AI | Advised by William Gantt Walden and Benjamin Van Durme

- Designed a reference-free Judge that audits report correctness and completeness using requirement graphs and a shared evidence ledger.
- Integrated rubric-guided refinement that targets decisive evidence gaps, recomputes feasibility, and resynthesizes reports while keeping the base model fixed.
- On 50 Gold-labeled claims, reduced score MAE from 1.12 to 0.88 and raised within-one-point accuracy from 70% to 82%.

Beijing Dajia Internet Information Technology, Beijing, CN 07/2023 - 03/2025
Software Development Engineer

- Built performance-analysis, monitoring, and traffic-replay tools for large-scale mobile systems, enabling rapid diagnosis of memory and performance issues.
- Maintained image, animation, and list-view components supporting large-scale events on Kuaishou.

TECHNICAL SKILLS

Programming: Python, C/C++, Objective-C, SQL, JavaScript
Systems & Data: Linux, Git, Slurm, Conda, Spark, Hive, PostgreSQL